Cancer Gene Expression Classification Using Machine Learning

Transcriptomics-Based Cancer Prediction Using GEO Gene Expression Data

Tech Stack:

Python
GEOparse
Scikit-learn
XGBoost
Gradient Boosting
Matplotlib
Seaborn

Problem Statement

Cancer diagnosis using transcriptomic data involves analyzing thousands of gene expression measurements, making manual interpretation difficult.

This project aims to develop a machine learning pipeline to classify samples as **Tumor** or **Normal** using gene expression data obtained from GEO.

Objective

Gene Expression Data
↓
Machine Learning
↓
Cancer Classification

Parameter	**Value**
Samples	296
Features	62,976
Classes	Tumor / Normal

Workflow

GEO SOFT File

↓

Expression Matrix

↓

Missing Value Handling

↓

Variance Filtering

↓

Scaling

↓

Train/Test Split

↓

Model Training

↓

Evaluation

Data Preprocessing

Preprocessing

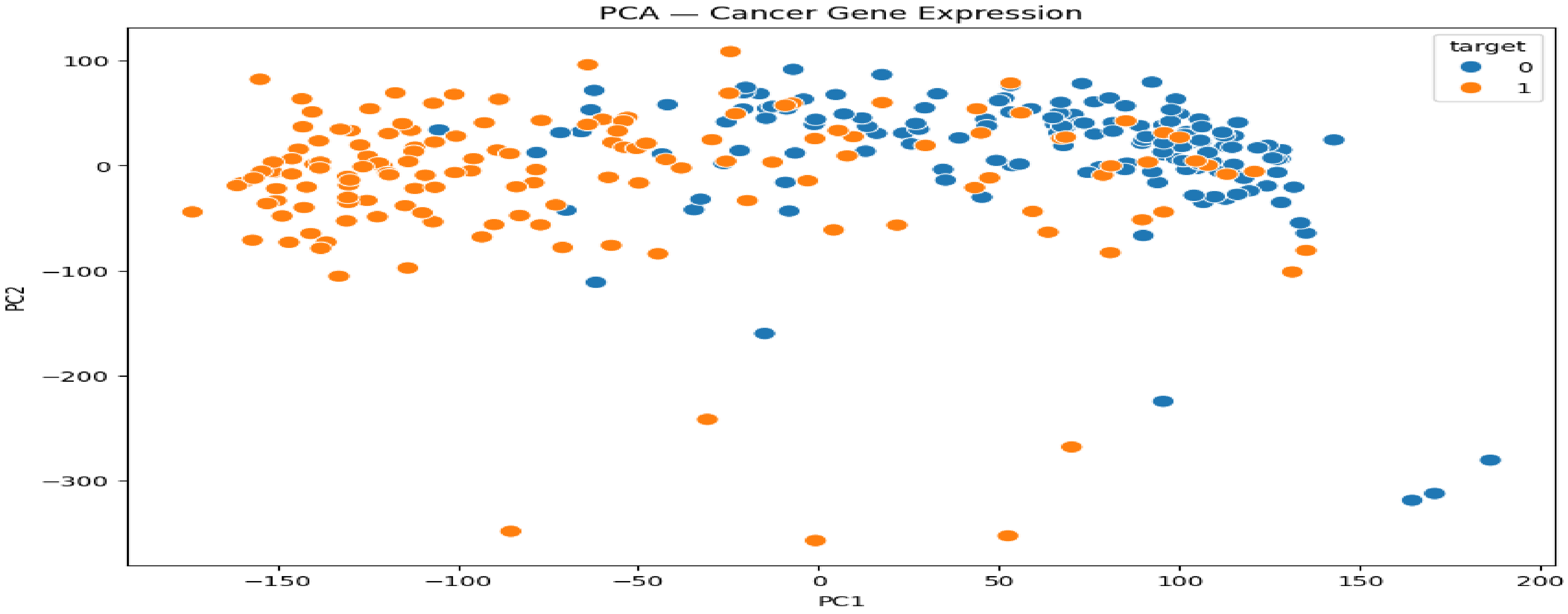
- Parsed GEO SOFT files
- Built expression matrix
- Missing value imputation
- Variance filtering
- Standardization

Final ML shape:

```
Expression Matrix Shape:
(296, 62976)

Dataset Shape:
(296, 62977)
```

Principal Component Analysis (PCA)



- PCA reduced high-dimensional gene expression into 2 components.
- Partial separation observed between classes.
- Class overlap indicates biological variability.

Observation

Gene expression patterns show sufficient separation to support cancer classification.

Model Comparison

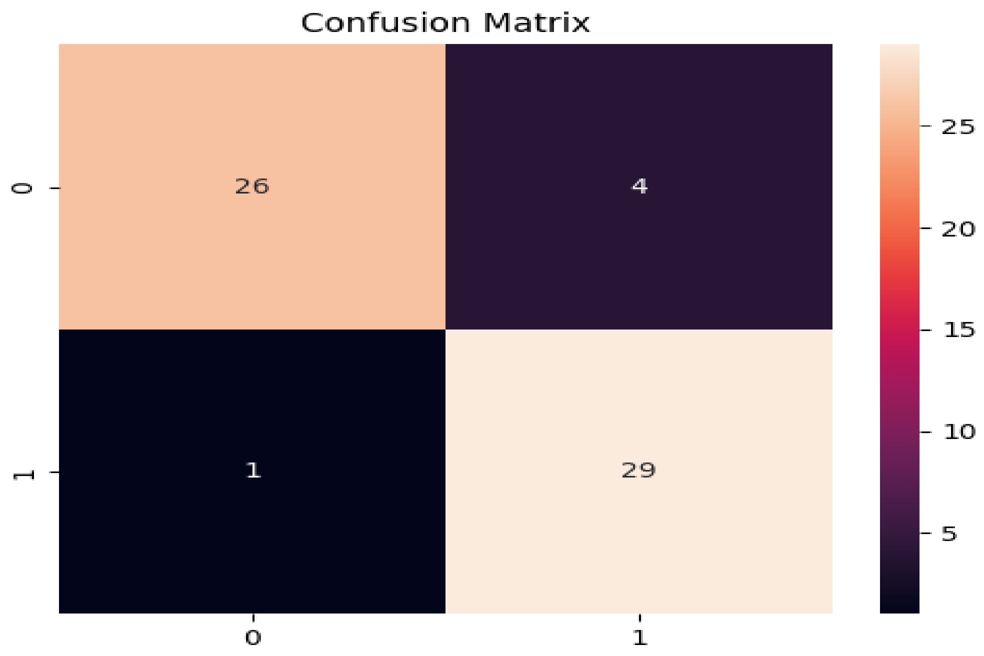
Metric	XGBoost	Gradient Boosting
Accuracy	**91.7%**	85.0%
Precision	**0.92**	0.86
Recall	**0.92**	0.85
F1 Score	**0.92**	0.85

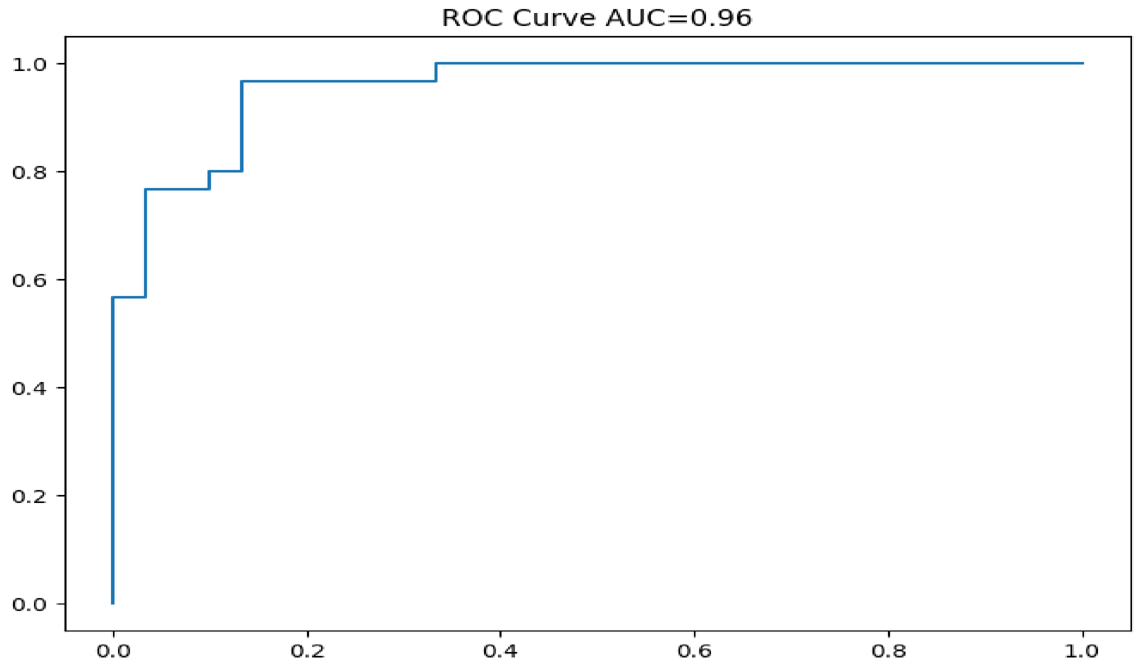
Observation

- XGBoost achieved higher overall performance
- Better precision and F1-score
- Selected as final model for cancer classification

Final Model:
XGBoost Classifier

Evaluation





Metric	**Value**
Accuracy	91.7%
ROC–AUC	0.96

Key Findings

Correctly classified **55 of 60 samples.**

Only **1 false negative** observed.

Strong discrimination capability (**AUC = 0.96**)

Final Model: XGBoost

Accuracy: 91.7%

ROC-AUC: 0.96

Conclusion

Outcome

- Built end-to-end transcriptomics ML pipeline.
- Converted GEO biological data into ML-ready format.
- Classified cancer samples successfully.
- Evaluated and interpreted results.

Repository & Tools

Tech Stack

- Python
- Pandas
- GEOparse
- Scikit-learn
- XGBoost
- Matplotlib
- Seaborn
- Jupyter Notebook

The Notebook of this project will be found on the my github profile.
GitHub:
Github

Notebook:
Analysis.ipynb